

Abhiram Mokkapati

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Education

Bachelor of Science in Electrical Engineering & Computer Science, Minor in Data Science Expected May 2027
University of California, Berkeley

Relevant Coursework: Principles & Techniques of Data Science, Efficient Algorithms & Intractable Problems, Discrete Math & Probability Theory, Computer Architecture, Data Structures & Algorithms, Computer Security

Skills

Programming: Java, Python, SQL, C, C++, HTML, JavaScript, Go

Frameworks: React, Next.js, Node.js, FastAPI, PyTorch, TensorFlow, LightGBM, NumPy, Pandas, OpenCV, MediaPipe, Vite

Tools & Platforms: Git, Docker, PostgreSQL, Supabase, AWS/Vercel, GitHub Actions, Socket.IO

Work Experience

Motion Performance CoPilot, McQueen.AI

San Francisco, CA

Data Engineer → Engineering Lead

Sept 2025 - Jul 2026

- Developed real-time stroke classification system using MediaPipe and OpenCV with temporal pattern analysis, achieving 75-85% accuracy across freestyle, backstroke, breaststroke, and butterfly, improving baseline detection by 15-25%.
- Engineered automated wall detection and lap tracking system leveraging direction-change algorithms and adaptive learning, enabling stroke counting, split time recording, and JSON session data export with 90%+ accuracy.
- Designed context-aware feedback engine integrating swimmer profiles and training set parameters to generate personalized coaching insights, comparing actual stroke counts and pace against intensity-based performance targets.
- Stepped up to Engineering Lead for an intensive summer sprint, architecting a GPU-accelerated multi-camera tracking pipeline (YOLOv8 + OSNet Re-ID) and a Next.js/Supabase coach-facing web platform, driving the majority of the codebase's growth during that period.

Projects

[Secure End-to-End Encrypted File Sharing System](#)

Go, Cryptography

- Designed and implemented a Dropbox-like client in Go with end-to-end encryption, supporting user authentication, file operations, cryptographically-signed sharing with RSA-2048, and secure revocation through file re-encryption at randomized locations.
- Engineered $O(n)$ efficient append operations using linked-list architecture, achieving ~1800 bytes constant bandwidth overhead regardless of file size, enabling seamless multi-gigabyte file updates without re-downloading entire files.

[Hall Staff Duty – Multi-Tenant RA Scheduling Platform](#)

Next.js, TypeScript, Supabase (PostgreSQL) – [GitHub](#)

- Architected a multi-tenant scheduling platform enforcing Postgres Row-Level Security across 4 role tiers, verified via manual impersonation testing to confirm zero cross-tenant data leakage.
- Built a custom greedy and local-search swap algorithm to auto-generate fair on-call schedules under availability and consecutive-day constraints, and shipped invite-only auth with rotating-token .ics calendar feeds.

F1 Race Outcome Prediction Model

Python, LightGBM, FastAPI, React

- Redesigned race-outcome prediction from independent per-target classifiers into a learning-to-rank model (LightGBM, LambdaRank/NDCG) producing race-relative driver orderings from historical FastF1 data.
- Validated a Plackett-Luce probability derivation that outperformed naive normalization on Brier score, and used time-based train/test splits on FastF1 data to prevent lookahead leakage into held-out seasons.